

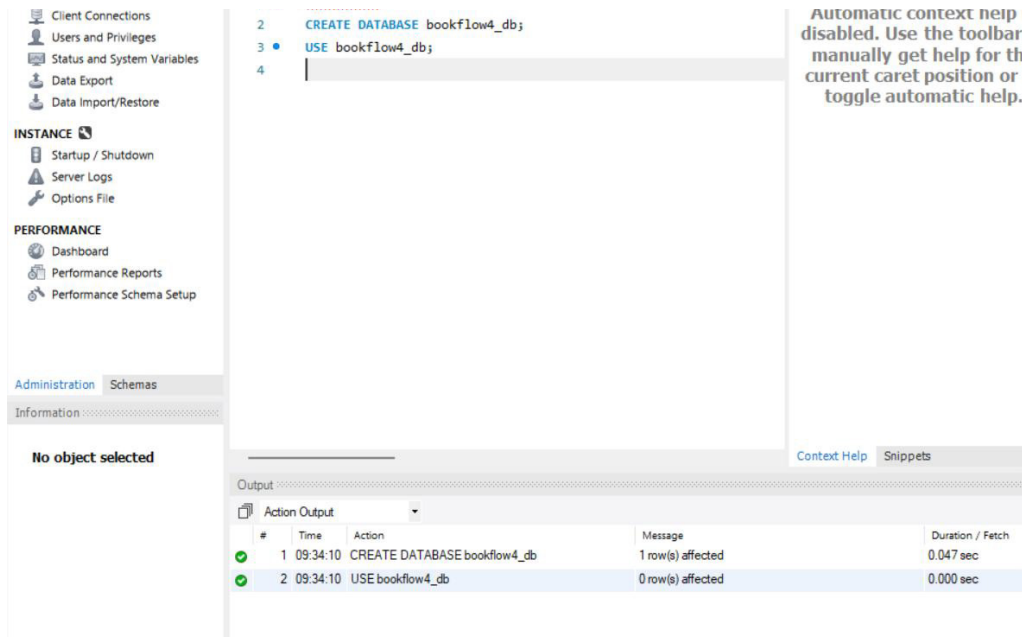
Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030414

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Query 1: -CREATE DATABASE



Automatic context help disabled. Use the toolbar manually get help for th current caret position or toggle automatic help.

Output

#	Time	Action	Message	Duration / Fetch
1	09:34:10	CREATE DATABASE bookflow4_db	1 row(s) affected	0.047 sec
2	09:34:10	USE bookflow4_db	0 row(s) affected	0.000 sec

Query 2: Create Books Table (DDL + Constraints)

```
CREATE TABLE BOK (  
    book_id INT PRIMARY KEY,  
    title VARCHAR(100) NOT NULL,  
    isbn VARCHAR(20) UNIQUE,  
    published_year INT CHECK (published_year < 2027)  
);
```

Output: -

✓	4	09:38:44	CREATE TABLE Bok (book_id INT PRIMARY KEY...	0 row(s) affected	0.109 sec
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Query 3: Insert Books and display

```
12 INSERT INTO Bok (book_id,title,isbn,published_year) VALUES
13 (1,'The Great Bay','9780743273565',2000),
14 (2,'The Indian Culture',2345987456,2010),
15 (3,'1947','9456783457934',2020);
16 SELECT * FROM Bok;
```

book_id	title	isbn	published_year
1	The Great Bay	9780743273565	2000
2	The Indian Culture	2345987456	2010
3	1947	9456783457934	2020
NULL	NULL	NULL	NULL

Output:-

✓	12	12:10:49	INSERT INTO Bok (book_id,title,isbn,published_y...	3 row(s) affected	Records: 3 Duplicates: 0 Wami...	0.062 sec
✓	13	12:10:49	SELECT * FROM Bok LIMIT 0, 1000	3 row(s) returned		0.000 sec / 0.000 sec

Query 4: Members Table: Create

```
CREATE TABLE Members(
member_id INT PRIMARY KEY,
full_name VARCHAR(100),
email VARCHAR(100) UNIQUE
);
```

Output:-

✓	19	12:20:46	CREATE TABLE Members(member_id INT PRIM...	0 row(s) affected	0.063 sec
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Query 5:- Insert Members table

```
INSERT INTO Members(member_id, full_name, email) VALUES
(1, 'Yishaani', 'yishaani@gmail.com'),
(2, 'Rahul Sharma', 'rahul@gmail.com'),
(3, 'Priya Reddy', 'priya@gmail.com');
```

Output: -

20	12:22:31	INSERT INTO Members (member_id,full_name,e...	3 row(s) affected Records: 3 Duplicates: 0 Wami...	0.016 sec
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Query 6: - : Display Members table

```
SELECT * FROM Members;
```

Output: -

102	Adhrittha	adhrittha@gmail.com
103	Manasvi	manasvi@gmail.com
*	NULL	NULL

INSERT INTO Members (member_id,full_name,e... 3 row(s) affected Records: 3 Duplicates: 0 Wami...
SELECT * FROM Members LIMIT 0, 1000 3 row(s) returned

Query 7: Create Loans Table

```
CREATE TABLE Loan (
load_id INT PRIMARY KEY,
member_id INT,
book_id INT,
loan_date DATE,
FOREIGN KEY (member_id) REFERENCES Members(member_id),
FOREIGN KEY (book_id) REFERENCES Bok(book_id)
);
```

Output:-

23	12:30:52	CREATE TABLE Loan (load_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIG...	0 row(s) affected
----	----------	---	-------------------

Query 8: Insert 10 Loans

```
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
```

Query 9: Display

```
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
SELECT * FROM Loans;
```

Output :-

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01
NULL	NULL	NULL	NULL

Query 10: JOIN Query

```
SELECT m.full_name AS Member_Name,b.title AS Book_Title
FROM loans l
INNER JOIN Members m ON l.member_id = m.member_id
INNER JOIN Bok b ON l.book_id = b.book_id;
```

Output: -

Book_Title
The Great Bay
The Indian Culture
1947
The Great Bay
The Indian Culture
The Great Bay
1947
1947
The Indian Culture
The Great Bay

103 13:50:44 SELECT m.full_name AS Member_Name,b.title A... 10 row(s) returned

Query 11: GROUP BY Query

```
SELECT published_year, COUNT(book_id) AS Total_Books
FROM Bok
GROUP BY published_year
ORDER BY published_year;
```

Output :-

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	published_year	Total_Books			
▶	2000	1			
	2010	1			
	2020	1			

✓	103	13:50:44	SELECT m.full_name AS member_name,b.title AS...	10 row(s) returned	0.000 sec / 0.000 sec
✓	104	13:54:20	SELECT published_year, COUNT(book_id) AS To...	3 row(s) returned	0.016 sec / 0.000 sec

Query 12: Donation via TRANSACTION

```
CREATE TABLE Donation_History (
donation_id INT PRIMARY KEY,
book_id INT,
donar_name VARCHAR(100),
donation_date DATE,
FOREIGN KEY (book_id) REFERENCES Bok(book_id)
);
```

Output: -

✓	107	13:57:39	CREATE TABLE Donation_History (donation_id I...	0 row(s) affected	0.094 sec
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Query 13:

```
START TRANSACTION;
INSERT INTO BOK (book_id,title,isbn,published_year)
VALUES (4,'Animal Farm','9978654321980',1945);
INSERT INTO Donation_History(donation_id,book_id,donar_name,donation_date)
VALUES(1,4,'Raj Kumar',CURDATE());
COMMIT;
```

Output: -

✓	109	14:02:19	COMMIT	0 row(s) affected
✓	110	14:02:25	START TRANSACTION	0 row(s) affected
✓	111	14:02:29	INSERT INTO BOK (book_id,title,isbn,published_...	1 row(s) affected

✓ 113 14:02:50 INSERT INTO Donation_History(donation_id,boo... 1 row(s) affected

Query 14: Index on ISBN

```
CREATE INDEX idx_books_isbn ON Bok(isbn);
```

Output :-

✓ 116 14:05:25 CREATE INDEX idx_books_isbn ON Bok(isbn) 0 row(s) affected Records: 0 Duplicates: 0 Wami... 0.125 sec

Query 15: Fast Search

```
SELECT * FROM BOK WHERE isbn = '9780743273565';
```

Output : -

	book_id	title	isbn	published_year
▶	1	The Great Bay	9780743273565	2000
*	NULL	NULL	NULL	NULL

✓ 117 14:07:50 SELECT * FROM BOK WHERE isbn = '97807432... 1 row(s) returned